Nathan Kap

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Education

Purdue University, West Lafayette, IN **Graduation:** May 2025
Bachelor of Science in Computer Engineering **GPA:** 3.0/4.0
Relevant Coursework: *Data Structures, Signals & Systems, Microprocessor Systems, Data Science.*

Work Experience

ManTech | Computer Engineer Intern *June 2023 - July 2023*

- Enhanced quality assurance for US Navy TI-16 encoders by developing an automated Python/SQLite testing solution.
- Collected and interpreted data to systematically validate outputs, identifying potential non-conformances and providing foundational data.

Technical Skills & Projects

Languages & Tools
C, Python, RISC-V Assembly, System Verilog, SQLite, MATLAB, Microsoft Office.

Purdue CREATE

- Developed embedded C firmware for an STM32-based motorized painting system, implementing safety interlocks to ensure system reliability and adherence to operational specifications.

IoT Backpack Anti-Theft Device

- Prototyped an ESP32-based (MicroPython) motion-activated security device, integrating an MPU6050 accelerometer.
- Implemented sensor calibration, motion detection algorithms, and Wi-Fi connectivity for remote activation (ThingSpeak) and real-time alert notifications (IFTTT).

Operating System Kernel Development (JOS)

- Developed core components of a microkernel-style OS, including virtual memory (paging, CoW fork), process management (scheduling, IPC), multiprocessor support, and a basic file system.
- Implemented system calls, interrupt handling, and x86 assembly for low-level operations and kernel design.

Community Leadership

Youth Executive Committee Board Member *Falam Baptist Church of Indiana | January 2022 - Present*

- Led a mentorship program for 10 mentors and 20+ mentees, fostering youth development.
- Forged a partnership with the international organization, Cru.

Sunday School Teacher & Music Director *Falam Baptist Church of Indiana | January 2022 - Present*

- Instruct and engage a class of 38 students weekly, developing and delivering educational content.
- Lead and mentor student musicians through rehearsals, preparing them for praise and worship services.

Project Coordinator *Hope For Tomorrow | January 2024 - Present*

- Founded and led a new international department for a non-profit organization.
- Coordinate with diverse stakeholders to facilitate collaboration and ensure project objectives are met.

Event Director *ChinTV International | September 2023 - Present*

- Produced a yearly awards event for international and local Burmese Chin artists and significant leaders by leading and coordinating a cross-functional team of over 20 volunteers in planning and logistics.